

Understanding Java Code Execution

Step-by-Step Walkthrough

Programming Tutorial

Code Execution Flow

```
2 package scripts;
```

```
public class LeapYear {
```

Code Execution Flow

```
package scripts;
```

3

```
public class LeapYear {  
    static boolean isLeap(int year) {
```

Code Execution Flow

```
package scripts;
```

```
4 public class LeapYear {  
    static boolean isLeap(int year) {  
        return (year % 4 == 0) && (year % 100 != 0) || (year %  
            400 == 0);  
    }  
}
```

Code Execution Flow

```
public class LeapYear {  
5  static boolean isLeap(int year) {  
    return (year % 4 == 0) && (year % 100 != 0) || (year %  
        400 == 0);  
}
```

Code Execution Flow

```
public class LeapYear {  
    static boolean isLeap(int year) {  
6      return (year % 4 == 0) && (year % 100 != 0) || (year %  
        400 == 0);  
    }  
}
```

Code Execution Flow

```
static boolean isLeap(int year) {  
    return (year % 4 == 0) && (year % 100 != 0) || (year %  
        400 == 0);  
}  
  
static void reportLeapYear(int year) {
```

Code Execution Flow

```
        return (year % 4 == 0) && (year % 100 != 0) || (year %  
            400 == 0);
```

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```
}
```

```
static void reportLeapYear(int year) {  
    boolean leap = isLeap(year);
```


Code Execution Flow

```
}
```

```
9
```

```
static void reportLeapYear(int year) {  
    boolean leap = isLeap(year);  
    System.out.println("isLeap(" + year + ")=" + leap);  
}
```

Code Execution Flow

```
static void reportLeapYear(int year) {  
10    boolean leap = isLeap(year);  
    System.out.println("isLeap(" + year + ")=" + leap);  
}
```

Code Execution Flow

```
static void reportLeapYear(int year) {
```

```
    boolean leap = isLeap(year);
```

```
11    System.out.println("isLeap(" + year + ")=" + leap);
```

```
}
```

Code Execution Flow

```
boolean leap = isLeap(year);  
System.out.println("isLeap(" + year + ")=" + leap);
```

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```
}
```

```
public static void main(String[] args) {
```

Code Execution Flow

```
        System.out.println("isLeap(" + year + ")=" + leap);  
    }
```

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```
public static void main(String[] args) {  
    int year = 2000;
```

Code Execution Flow

```
}
```

```
14
```

```
public static void main(String[] args) {  
    int year = 2000;  
    reportLeapYear(year);
```

Code Execution Flow

```
public static void main(String[] args) {  
    int year = 2000;  
    reportLeapYear(year);  
}
```

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Code Execution Flow

```
public static void main(String[] args) {  
    int year = 2000;  
    reportLeapYear(year);  
}
```

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Code Execution Flow

```
int year = 2000;  
reportLeapYear(year);
```

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}